MCS-201	ADVANCE DATABASE SYSTEMS	L T P 3 1 0	4 Credits

Course Objectives:

CO1	Understand the role of a database management system in an organization.
CO2	Understand basic database concepts, including the structure and operation of the relational data model.
CO3	Construct simple and moderately advanced database queries using Structured Query Language (SQL).
CO4	Understand the concept of a database transaction and related database facilities, including concurrency control, journaling, backup and recovery, and data object locking and protocols.
CO5	Design and implement a small database project using Microsoft Access

MODULE I

Transaction and schedules, Concurrent Execution of transaction, Conflict and View Serializability, Testing for Serializability, Concepts in Recoverable and Cascade-less schedules. Lock based protocols, time stamp based protocols, Multiple Granularity and Multi-version Techniques, enforcing serializablity by Locks, Locking system with multiple lock modes. Distributed Transactions Management.

MODULE II

Data Distribution, Fragmentation and Replication Techniques, Distributed Commit, Distributed Locking schemes. Issues of Recovery and atomicity in Distributed Databases, Traditional recovery techniques, Log based recovery, Recovery with Concurrent Transactions, Recovery in Message passing systems, Checkpoints, recovery line.

Distributed Query Processing, Multi-way Joins, Semi joins, Cost based query optimization for distributed database, Updating replicated data, protocols for Distributed Deadlock Detection, Eager and Lazy Replication Techniques

MODULE III

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Object-relational databases, active databases, and multi-databases. Overview of modern database technologies, such as parallel databases, multimedia databases, spatial and temporal databases, data warehousing and data mining, deductive databases.

References

1. *Silberschatz, Korth and Sudershan, "Database System Concept", Mc Graw Hill.*
2. *Ramakrishna and Gehrke, "Database Management System", Mc Graw Hill.*
3. *Garcia-Molina, Ullman,Widom, "Database System Implementation" Pearson Education.*
4. *Ceei and Pelagatti,"Distributed Database", TMH.*
5. *Singhal and Shivratri, "Advance Concepts in Operating System" MC Graw Hill.*

Course Outcome :

CO1	Describe the fundamental elements of relational database management systems
CO2	Explain the basic concepts of relational data model, entity-relationship model, relational database design, relational algebra and SQL
CO3	Design ER-models to represent simple database application scenarios
CO4	Convert the ER-model to relational tables, populate relational database and formulate SQL queries on data
CO5	Familiar with basic database storage structures and access techniques: file and page organizations, indexing methods including B tree, and hashing.